Predictive structure emerges during generalisation of kin terms to new referents

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Abstract

Despite crosslinguistic diversity in how kin relations map to terminology, there are constraints on which kin may be categorised together. But what are the constraints on kin term variation, and where do they come from? One proposed constraint is internal co-selection — an evolutionary process where terminological changes in one generation of kin co-occur with parallel changes in other generations. This results in kin terms which are predictable on the basis of other kin terms, a property we call predictive structure. To determine the strength of this constraint, we measured the predictive structure of systems of kinship terminology from 1135 languages. We found that kinship systems exhibit a significant degree of predictive structure, and we argue that its prevalence reflects a pressure for simplicity. We expected that this pressure would be imposed during the generalisation of known kin terms to new referents, and we tested this claim using an artificial kin term generalisation task. Our results suggest that generalisation of kin terms does indeed favour predictive structure, but that people's preference for predictive structure faces interference from other pressures to distinguish kin by features like gender.

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1. Introduction

When we acquire language, we learn how words map to objects or states of affairs in the world around us. In doing so, we determine what the plausible semantic contrasts are in our language. Different languages reach different solutions to this problem of categorisation, as we know from cross-linguistic investigations of semantic domains such as colour (Berlin & Kay, 1969; Regier et al., 2007), number (Greenberg, 1978; Y. Xu et al., 2020), spatial relations (Levinson & Meira, 2003; Majid et al., 2004; Khetarpal et al., 2009), and the focus of this paper, kinship (Murdock, 1949; Greenberg, 1966, 1990; Kemp & Regier, 2012). Consider the English kin term *uncle*, for example. An English speaker can use *uncle* to refer to any of their parents' male siblings, while a Hindi speaker has three distinct terms for the same group of kin: *māmā* 'mother's brother', *tāū* 'father's older brother', and *cācā* 'father's younger brother'. Meanwhile, in Hawaiian, the term *makuakane* includes all of the male siblings of both parents and also one's own father. In spite of surface variation in the way kin are partitioned across different languages, the extent of the possible variation is highly constrained. Some theoretically possible patterns are never attested at all. There is no language, for instance, with a single term for grandfathers and brothers, or for nieces and grandsons. In this paper, we address the extent of these constraints on kinship variation, and propose a mechanism by which they emerge and persist.

1.1. Proposed constraints on variation

Anthropologists have sought to explain the pattern of constrained variation with a small set of featural constraints such as gender distinctions, generational distinctions, or relative age (seniority) distinctions (Kroeber, 1909; Fox, 1979), and a more recent body of work situates these featural constraints in a hierarchical framework (Hage, 1999; Jones, 2010). Perhaps we cannot have 'niece' and 'grandson' mapped to a single term simply because they differ along dimensions of generation and gender? While tempting, such a solution cannot capture the scope of the variation in distinctive features we see cross-linguistically. For instance, Omaha categorises kin together from different generations with the term *e-nä'-hä*, meaning 'father', 'father's brother', and 'father's brother's son'. In Indonesian, the term *kakak*, meaning 'older brother' and 'older sister', categorises kin of varying genders together. Furthermore, kinship systems rarely make the same distinctions across the entire paradigm: in English, our grandparents, parents, siblings, children and grandchildren are all distinguished by gender, but not our cousins. And even if we were satisfied by the descriptive power of a feature-based explanation, we would still lack an explanatory account of the mechanisms that give rise to the variation we observe in kinship terminologies.

Elsewhere in the cognitive sciences, the pattern of constrained variation in kinship systems has been explained using principles of efficient communication (Kemp & Regier, 2012; Kemp et al., 2018). Systems of kinship terminology are constrained by an optimal trade off between opposing pressures to be simple and expressive; systems that fail to trade off these pressures efficiently are unattested. Kinship terminology systems are simpler if they have fewer terms and if these terms have simple, compressible meanings - disjunctive categories like

'a sibling who is older than you *or* male' are dispreferred compared to contiguous categories like 'sibling who is older than you *and* male' (Nerlove & Romney, 1967). On the other hand, kinship terminology systems are more expressive if the probability of correctly interpreting the intended referent of a kin term is maximised. This means that expressive kinship systems have more terms that each denote fewer individuals, and high frequency referents have more precise terms, reducing the chance of a term being misunderstood. This account explains why we have a wide range of surface variation, but also defines clear boundaries on the extent of that variation: for a given level of communicative precision kinship terminology systems are as compressible as possible, and for a given level of compression they are as precise as possible. In short, we shouldn't expect to find kinship systems that are both complex and highly ambiguous.

Here we seek to understand the scope of diversity in kinship terminology systems from an evolutionary perspective, in tandem with the mechanisms that give rise to that diversity — an under-researched yet rich union between language, culture and cognition (for a review, see Mitchell & Jordan, 2021). Constraints on language variation are not random. They reflect adaptations to the cultural transmission of language, such that simpler structures that are more learnable and expressive structures that are more useful for communicating proliferate (Kirby et al., 2008; Kirby et al., 2015; Smith, 2022). We can reformulate the trade-off between simplicity and expressivity in semantic category systems as a joint adaptation to language learning and communication (Carr et al., 2020). Indeed, crosslinguistic constraints on colour terminology have been associated with biases imposed during cultural transmission (J. Xu et al., 2013), and cultural transmission should similarly be the process by which kinship terminology is

optimised for efficient communication (Kemp & Regier, 2012). Previous work offers some insight on the role of learning and communication in the design of kin terms: child language acquisition studies find that semantically simpler kin terms like *father* and *brother*, which require understanding only a single relation, are learned earlier than those like *uncle*, which requires additionally that a child understands both *father* and *brother* (Haviland & Clark, 1974). Experimental work finds that simpler kinship systems (where kin are categorised by shared features) are more learnable by adults in the lab than those where kin terms are shared by less similar individuals (Smith et al., 2020). Additionally, corpus work shows that kinship terms are optimised for the communication needs of individuals, reflecting the relative frequencies with which particular kin are referenced (Anand & Regier, 2023).

1.2. Internal co-selection

What common features might we then expect systems of kinship terminology to have, if they are constrained by opposing pressures to be simple and expressive? One proposed structural constraint on kinship systems is *internal co-selection* (Passmore et al., 2021): a process of kin term change where new distinctions or mergers in one part of the kinship system are followed by parallel changes in another part of the system. Figure 1 displays an example of this process. If a merger between the meanings 'father' and 'uncle' results in a single term *funcle* for both referents, we might find ourselves in the confusing situation where sometimes the child of a *funcle* is called our *brother* and sometimes they are called our *cousin*. Choosing the appropriate term becomes a difficult task, since we cannot rely on our knowledge of other kin terms to arrive at the correct term. We should therefore expect a second merger between brother and

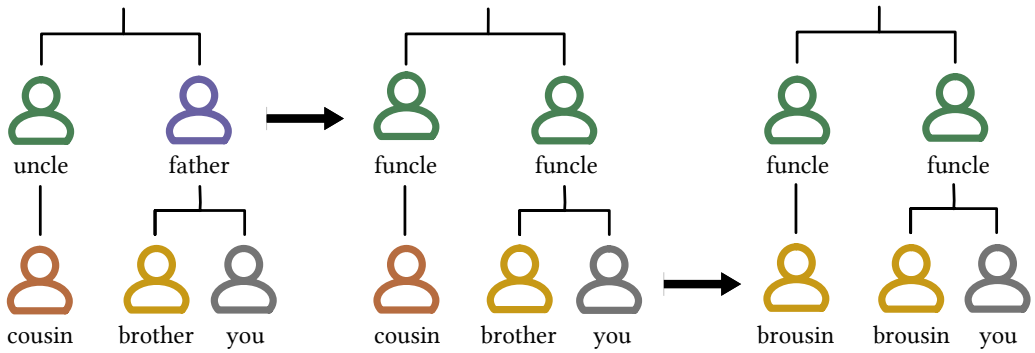


Figure 1: An example of internal co-selection. The system on the left undergoes a merger, such that the previously unique terms *father* and *uncle* collapse to a single term *funcle* which can be used for both referents. Under the internal co-selection hypothesis, we expect a parallel merger in the terms for their children, as it is no longer clear when the child of a *funcle* should be a *brother* or a *cousin*. By merging these two terms to a single term *brousin*, we remove this complexity from the kinship system (at the cost of reduced expressivity).

cousin to a single category *brousin*. Then, the child of a *funcle* is always called a *brousin*, and we can rest easy knowing that we do not have to remember complex information about what our *funcle*'s child is called under what particular circumstances.

If kinship systems evolve via linked sets of changes, then we get constraints on possible relationships within our kinship system 'for free': if we have a single term for fathers and uncles, the terms for their children cannot vary arbitrarily, but also become constrained to a single term. We refer to this property as the **predictive structure** of the kinship system. The more accurately you can predict terms for individuals in one generation (e.g. sibling and cousin terms) from terms for individuals in another generation (e.g. terms for parents, aunts and uncles), the more predictive structure the system has. We propose that predictive

structure is one method by which kinship terminologies optimise for efficient communication, as predictive structure facilitates greater simplicity. Predictive kinship terminologies are more compressible, since each kin term can be defined with respect to other terms (e.g. we use our understanding of the term *funcle* to define the term *brousin*).

If internal co-selection is a driving force in the evolution of kinship terminology, predictive structure should be common in systems of kinship terminology cross-linguistically. However, Passmore et al. (2021) did not find consistent evidence that kinship systems are structured predictively – at least, not in the ways presumed by common organisational typologies for kinship systems (Murdock, 1949). In Sections 2 – 3, we extend their study using an alternative, information-theoretic measure of predictive structure in order to robustly test the prediction that kinship terminology systems are structured predictively. We present the broadest typological analysis of the structure of kinship systems to date, finding that the world's kinship systems are structured more predictively than we would expect by chance (if there were no evolutionary pressures imposing this structure). By offering a common structural principle that underlies simplicity in kinship systems, we support the claim that kinship systems are optimised for efficient communication.

What then would be the mechanism that facilitates internal co-selection as a process of kin term evolution? Why would kin terms evolve via linked sets of changes? Inspired by explanations for predictive structure in other domains such morphological paradigms (Ackerman & Malouf, 2013; Johnson et al., 2021), we propose that pressures imposed during cultural transmission incentivise predictive structure. Individuals faced with incomplete input when learning kin terms

will predict unknown terms from their existing knowledge: the children of *funcles* sharing a term seems a likely hypothesis, so upon hearing that one *funcle*'s child is a *brousin*, the learner extends that term to children of all *funcles*, regularising the paradigm. In light of the paucity of experimental approaches in kinship studies, we offer an empirical test of whether the generalisation process presupposed by the internal co-selection hypothesis does indeed give rise to predictive structure. In Section 4 we present the results of an artificial language generalisation task, showing that predictive structure emerges during generalisation of known kin terms to new referents.

2. Study 1: How common is predictive structure?

If kinship terminology evolves by internal co-selection, we would expect that kinship terms in one part of the system will be predictive of other parts of the system. We test this claim by comparing kinship terms across two generations: one's siblings and cousins (**Ego's generation** or **G0**) and one's parents and their siblings (**Ego's parents' generation** or **G+1**). Previous work has not found consistent evidence that kinship systems exhibit predictive structure (Passmore et al., 2021), but here we conduct a more robust test using a larger quantity of data from a greater variety of languages, and using a richer measure of predictive structure.

Passmore et al. (2021) measure predictive structure using a binary measure inferred from the typology for kinship systems put forward by Murdock (1949), which classifies kinship systems into six categories depending on the way a language distributes kin terms for cousins. In their study, if the organisation of kin at G+1 is classified the same as kin at G0, then that kinship system scores

1 (predictive structure). If the two categories differ, the language scores 0 (no predictive structure). However, this measure assumes that the ideal state for kinship systems following the internal co-selection process is one of Murdock's six types, which do not necessarily exhibit predictive structure. Additionally, a binary measure cannot capture the extent to which a system exhibits predictive structure given other constraints, such as the number of terms or what other distinctions are made.

Furthermore, while Murdock's typology is still commonly used in kinship research as a measure of kin term variation (Cronk et al., 2019; Trautmann & Whiteley, 2012) or to infer social kin organisation (Schulz et al., 2019), it is increasingly perceived as an inadequate and outdated description of global kinship variation (Passmore & Jordan, 2020). Indeed, Passmore et al. (2021) suggest that their inconclusive result reflects that the typology fails to accurately capture the extent of global variation.

Here, we use a new information-theoretic measure for predictive structure. We calculate predictive structure as the mutual information between kin terms, and use Monte Carlo methods to compare each kinship system to a null baseline, to measure the extent to which systems exhibit predictive structure relative to the predictive structure we would expect to occur by chance in such a system.

2.1. Data

Our data on the world's kinship systems comes from Kinbank (Passmore et al., 2023), a database of kinship terminology in 1215 languages across 21 major language families. Kinbank contains terms for relatives spanning five generations from grandchildren to grandparents.

For our analysis, we look only at kin terms for individuals in G0 including

siblings (elder and younger) and parents' siblings' children, and in G+1 including parents and parents' siblings (elder and younger). Taking into account gender, this leaves us with a total of 60 types of relative across both generations.

Limiting our analysis to two generations aligns with Passmore et al.'s (2021) claim about internal co-selection, which only considers the evolutionary link between G0 and G+1. Additionally, these relationships are the most reliably documented across the dataset, which allowed us to maximise the size of our data sample. When data was missing, we calculated the predictive structure of the terminology that was available, providing there was at least one term listed for a person in each generation. This criterion left us with data from 1135 kinship systems for our analysis.

In cases where more than one term was recorded for a given relative, we took the first listed term for our analysis. If two or more alternative spellings were recorded for a kin term, we took the first listed spelling.

2.2. Measuring predictive structure

For our analysis, we needed a robust measure of predictive structure and a baseline model against which to compare natural languages. The baseline should capture our expectations under the null hypothesis - i.e. that if there were no evolutionary pressure for predictive structure, kinship terminologies would not be structured in this manner.

Mutual information. We measure predictive structure in information theoretic terms, as the mutual information between generations of kin terms. Between two terms, mutual information tells us how much information we can obtain about a term T_0 from observing a term T_1. If we sum over all pairs (T_0, T_1) with

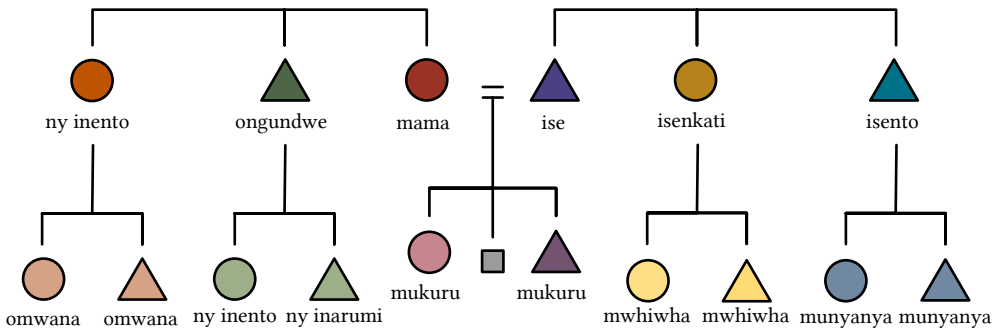


Figure 2: The kinship system of Nyoro. All terms are given relative to Ego, represented by the grey square. Circles represent women, triangles represent men. Colours indicate a unique kin term. Nyoro has a high degree of predictive structure: terms for Ego's siblings and cousins are uniquely specified by terms for Ego's parents and parents' siblings (nuncles). For instance, the child of a *ny intento* is always a *omwana*, never one of the other possible terms for a cousin. Note that Nyoro exhibits generational skewing, so the term *ny inento* can refer both to a mother's sister and a mother's brother's daughter. Nonetheless, it exhibits predictive structure as parent terms and child terms are uniquely paired.

T_0 in G0 and T_1 in G+1, we get a single score for each language that tells us how well we can predict the terms in G0 knowing only the terms in G+1 (and vice-versa). The mutual information I between two generations G0 and G+1 is given by:

$$I(G_0; G_{+1}) \equiv H(G_0) - H(G_0|G_{+1}) \tag{1}$$

where $H(G_0)$ is the entropy (in bits) over all kin terms in Ego's generation and $H(G_0|G_{+1})$ is the conditional entropy over all kin terms in Ego's generation given the terms in Ego's parents' generation. $H(G_0) = 0$ if a language uses a single term for all relatives in G0. Higher entropy values reflect a greater

number of terms, particularly if the distribution of terms is relatively uniform. $H(G_0|G_{+1}) = 0$ if the kin terms for relatives in G0 are completely predictable by knowing the term for each relative's parent in G+1. For a given number of distinct terms, $MI(G_0; G_{+1})$ is maximised when every term in G0 is uniquely specified by a term in G+1, as seen in Nyoro (Figure 2). $MI(G_0; G_{+1})$ is minimised when there is no way to reliably predict a term in the variable generation, i.e. when there is no variation in G0 terms. In reality, most languages sit between these extremes, but we predict that they will nonetheless have greater mutual information than we would expect if their kin terms were distributed across the space of possible referents in the absence of any pressure for predictive structure.

A baseline with no bias for predictive structure. We used Monte Carlo techniques to test whether given level of predictive structure is likely to arise by chance, comparing each language's true mutual information score to a null baseline with no bias for predictive structure, but with all other features of the language held constant. For each language, we generated baseline kinship systems by generating random permutations of the kin terms in G0, randomising which kin terms were associated with which referents, and thereby breaking any predictive bonds between terms in G0 and terms in G+1 while preserving the number of terms in each generation. Figure 3b exemplifies this manipulation for the Kurdish kinship system.

For each of the 1135 natural kinship systems in our dataset, we generated 1000 such permutations and measured the mutual information of each permutation. We compared the resulting distribution of theoretically possible mutual information values to the mutual information of the veridical kinship system. We calculated a z-score for each language using the mean and standard devia-

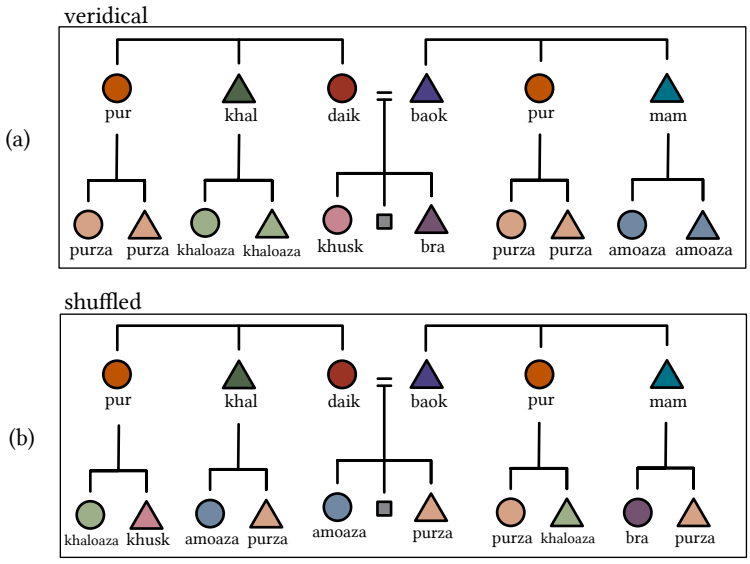


Figure 3: (a) The kinship system of Kurdish. Tree conventions as in Figure 2. Kurdish has a high degree of predictive structure, but these kin terms are also compositional: e.g. *pur* and *purza* share part of their form. (b) An example permutation of Kurdish. This hypothetical kinship system has low predictive structure: there is no reliable way to predict what a child will be called on the basis of what their parent is called. Before, the child of a *pur* was a *purza*; now the child of a *pur* might be *amoaza*, *bra*, *purza* or *khusk*.

tion of the baseline distribution, indicating how extreme the observed predictive structure is relative to what would be expected if there were no bias for predictive structure. We classified a kinship system as having a significant degree of predictive structure if it had higher MI than 95% of its permutations (i.e. a significance threshold of $p = 0.05$), measured as a mutual information value more than 1.96 standard deviations from the baseline mean ($z > 1.96$).

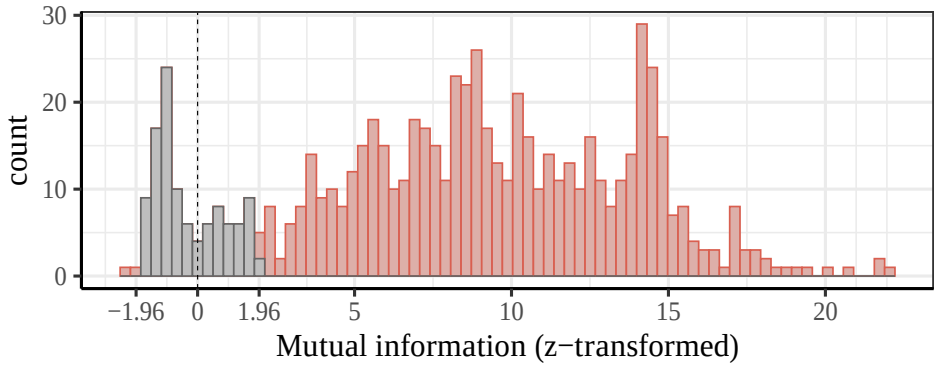


Figure 4: Distribution of mutual information z-scores for all languages in our dataset. Red bars indicate languages with significant mutual information values ($p < 0.05$). Dashed line shows $z = 0$. The vast majority of languages exhibit a significant degree of predictive structure, with the majority beating their baseline by an extremely large margin of > 5 standard deviations.

2.3. Results

We expected that natural systems of kinship terminology would exhibit a high degree of predictive structure, following the prediction from Passmore et al. (2021) that kin terms evolve via a linked set of changes. In terms of our measure, we expected that natural languages will have greater mutual information than 95% of their permutations.

Comparing the mutual information score for each natural language to the distribution of mutual information scores in its bespoke baseline, we found that the vast majority of languages in our dataset (87.5%) had significantly higher mutual information scores than their null baselines (see Figure 4). Only two (0.02%) languages had mutual information values significantly lower than their baseline; the remaining 12.5% of languages' mutual information was in the range we would expect by chance if there were no bias for predictive structure ($-1.96 < z < 1.96$).

Figure 5 shows the results for a sample of languages in our dataset. For al-

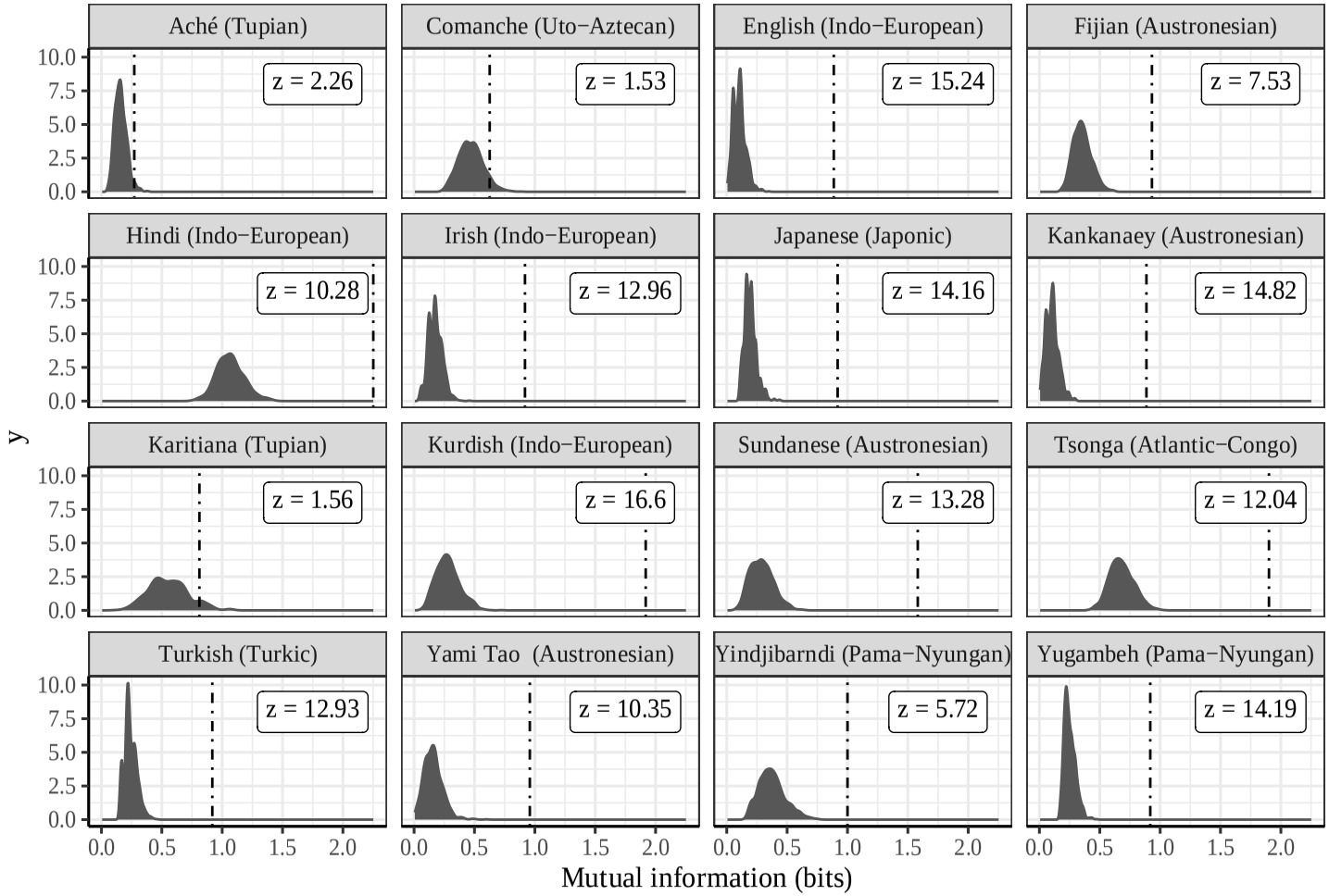


Figure 5: Range of mutual information values for a sample of languages in our dataset. Distributions represent the baseline mutual information from each of the 1000 permutations of each system; dotted lines represent the mutual information of the veridical kinship system. Labels give the z-score for each language - *z* greater than 1.96 indicates a significant degree of mutual information. In almost all cases, the true mutual information values far exceeds the bulk of the baseline distribution, indicating that systems of kinship terminology are structured more predictively than expected if there was no bias for predictive structure.

most all of these languages mutual information values for the natural languages were significantly higher than the baseline distribution, and often by a considerable margin, with many languages surpassing more than five standard deviations from their baseline mean.

2.4. Discussion

In accordance with our hypothesis, we found that kinship systems have much greater predictive structure than we would expect if there were no bias for predictive structure. Compared to randomly permuted baselines, we found that natural kinship systems exhibit vastly more structure; there is often not a single permutation out of 1000 that approaches the true mutual information value. This confirms quantitatively that natural language kin terms are not randomly allocated to referents, and further suggests that systematic predictive structure is a near-universal property of kinship systems.

We found some languages whose mutual information was not significantly different from their permuted counterparts. Our analysis was very conservative, and we included languages that had as few as a single term in each generation recorded in Kinbank. Languages whose mutual information fell within the set of values in the baseline were often those with incomplete kin term data (see Appendix 1), resulting in baselines with only a handful of possible permutations. We expect that had the data been complete, the baselines for these languages would have been more variable, and that their veridical mutual information would likely be more different from chance than we see here.

Our results contrast with the findings from Passmore et al. (2021), who did not find consistent evidence that kinship systems evolve by internal co-selection. By measuring predictive structure as the mutual information between two genera-

tions of kin terms, we were able to show the extent to which a particular kinship system exhibits predictive structure, while Passmore et al.'s (2021) measure was binary - either kin terms were predictive or they were not. By comparing our results to a null distribution, we were able to measure for each language how extreme the degree of predictive structure was. Additionally, we used a larger quantity of data from a greater variety of languages. We are confident that the results we have presented here are a stronger test of predictive structure.

3. Study 2: is predictive structure due to compositionality?

An alternative explanation for our findings in Section 2 could be that high mutual information is a by-product of compositionality, the pervasive feature of languages whereby the meaning of an expression is composed of the meanings of parts of that expression. It is not uncommon for kin terms to be compositional: morphologically transparent such that parts of their form are associated with parts of their meaning. For instance, Kurdish *khal* 'father's brother' and *khaloaza* 'father's brother's child' share part of their form, such that *khal* indicates which branch of the family tree these kin belong to - i.e. the father's brother's branch (see Figure 3a). Additionally, *-oaza* indicates that an individual is a child, and is also present in the other cousin terms *pur-za* and *am-oaza*. Labels which are compositional in this way inevitably have a high degree of predictive structure.

Compositionality has itself been argued to be in part a product of pressures for efficient communication (Kirby et al., 2015). Could the predictive structure we see in Section 2 be due entirely to compositionality? If so, it would no longer be clear that predictive structure emerges due to a specific pressure for kin terms in G+1 to uniquely specify kin terms in G0 - instead, this could be a side effect

of a more general preference for compositional structure in language.

Many of the kinship systems we see that have high mutual information relative to their baselines do not have clearly compositional kin terms - e.g. Nyoro in Figure 2, and of those in Figure 5, English, Fijian, Turkish, Kankanyaey, Yami, and Yugambeh have no clear compositional structure. But in order to more rigorously rule out the possibility that high mutual information is merely compositionality in disguise, in this section we measure the compositionality between terms denoting a parent and child pair (e.g. *uncle* and *cousin* in English) across our sample of kinship systems, and compare this to the measure of predictive structure for each language as computed in Section 2.

3.1. Data

The data used in this analysis are identical to those described in Section 2.1.

3.2. Measuring compositionality

As a proxy for compositional structure between parent and child kin terms, we measured the normalised Levenshtein distance (Levenshtein, 1965) between pairs of terms: the number of single-letter insertions, deletions, or replacements needed to turn one string into another, proportional to the length of the longer string. For instance, the number of changes required to change the term *uncle* into the term *cousin* is 6, as none of the letters are in the same position and one letter must be added. The length of the longer word *cousin* is 6, so the normalised edit distance between these terms is 1. In other words, this is the highest possible edit distance between a pair of words. To take an example from a language that does exhibit compositionality, consider Kurdish *pur* 'mother's sister' compared to *purza* 'mother's sister's daughter' - the normalized edit distance between these

terms is 0.4. The smaller the normalised edit distance, the more similar the two strings, with zero indicating that the strings are identical. In a kinship system that is compositional in the same way as Kurdish, with terms for offspring being related to terms for their parents, the edit distance between parent and child terms will tend to be low.

We computed the average edit distance between all pairs of parent and child terms. As in Section 2, we built null baselines for each kinship system by permuting which G0 terms were associated with which referents and measured the new average edit distance across the new set of parent-child pairs generated by this process, calculating z-scores for the edit distance of the true kinship system with respect to this random baseline. Kinship systems were considered significantly compositional if the true edit distance was smaller than 95% of its permutations, i.e. a z-score less than 1.96.

3.3. Results

Figure 6 shows the average edit distance z-score for each natural language. We found that languages tend not to have a significant degree of compositionality with respect to their null baselines; 50% of languages are non-compositional, while only 30% of languages have a significantly low average edit distance (i.e. are highly compositional). Given that extreme values of predictive structure are much more common (87.5%), this suggests that predictive structure is not merely an artifact of compositionality.

Additionally, we also found that 20% of kinship systems have edit distances that are significantly *greater* than those in the baseline, meaning that the labels for corresponding pairs of G+1 and G0 kin (i.e., parents and their children) are less similar to each other than they are to other kinship terms in the system.

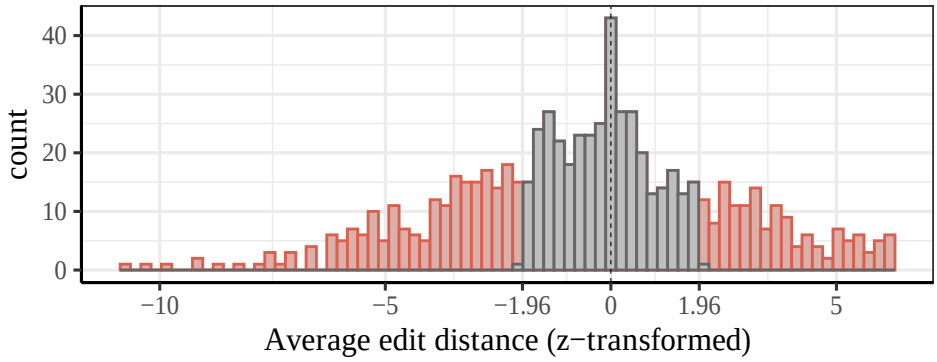


Figure 6: Distribution of average edit distance z-scores for all languages in our dataset. Since compositional systems have lower edit distances, *negative z-scores* indicate greater compositionality (the average distance between terms in the system is *less* than chance). Red bars indicate languages whose average edit distance beats a significance threshold of $p < 0.05$; grey bars indicate languages which do not have significant average edit distance. Dotted line shows $z = 0$. A large proportion of languages (50%) are not compositional, while predictive structure is more common (87.5%), suggesting that it is possible to have a high degree of predictive structure without compositionality in kin term forms.

Our edit distance measure is highly specific to the cross-generational predictive structure we identified in Study 1, only measuring the similarity between terms for a parent and their offspring. However, this is not the only manner in which languages encode meaningful information in kin terms. For instance, Figure 8 shows the kinship system from Andi, where the kin terms for 'brother' and 'father's brother' share a chunk of their form, *-woc̄i*. This morpheme encodes a meaning like 'this person is a male sibling'. Because of this similarity in form, our edit distance measure would give a higher score to Andi if the term *woc̄i* was used for 'father's brother's son'.

To confirm that predictive structure is not associated solely with composi-

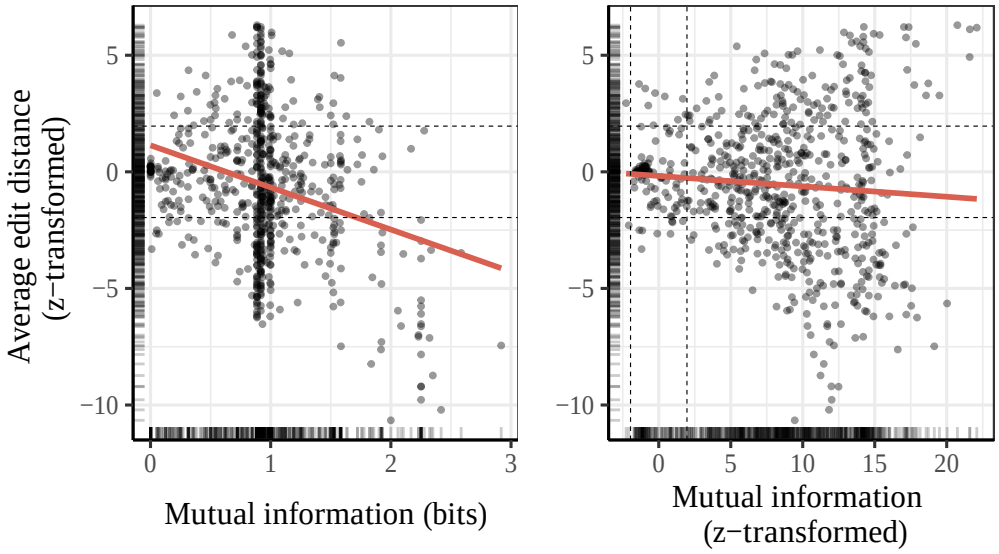


Figure 7: Left: Relationship between z-transformed average normalised edit distance and mutual information. There is a small inverse effect: as edit distance decreases (i.e. terms are more similar), mutual information increases. Right: Relationship between z-transformed average normalised edit distance and z-transformed mutual information. Dashed lines indicate significance thresholds. Extremely low edit distance is not correlated with extremely high mutual information, suggesting that predictive structure is not driven by compositionality.

tionality, we compared the z-transformed average edit distance between parent-child terms with the mutual information score for each language. 30% of languages in our sample have significantly lower average edit distance than chance, while 50% are at chance. Compositionality is not as common as predictive structure, but it is still prolific cross-linguistically. Figure 7 depicts the relationship between these variables: as kin terms become more compositional, predictive structure increases, as we expected. However, more crucially we found that among the 50% of kinship systems who have edit distance values that are indistinguishable from the baseline, 73% have mutual information values that dif-

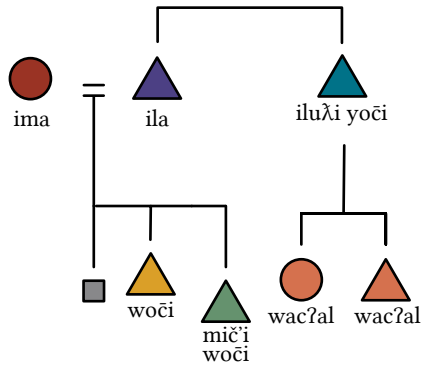


Figure 8: Truncated kinship system for Andi (Nakh-Daghestanian) including terms for parents, male siblings (elder and younger), father's brother, and father's brother's children. Tree conventions as in Figure 2. Longer lines indicate younger siblings, shorter lines indicate older siblings. Andi has extremely high average edit distance according to our measure because terms that share parts of their form are not parent-child pairs: *iluʎi woc̄i* 'father's brother' shares a chunk of its form with *woc̄i* 'elder brother' compared to *wacʔal* 'cousin'.

fer significantly from the baseline, suggesting that predictive structure need not co-occur with compositionality.

Importantly, the majority of systems with significantly high edit distance values nevertheless still have significantly high mutual information values (87.5%). For instance, Andi has a edit distance z-score of 5.98, and a mutual information z-score of 12.87 ($p < 0.001$), further supporting our claim that predictive structure is not merely reflective of compositionality.

3.4. Discussion

While predictive structure and compositionality are intrinsically related, our results suggest that predictive structure is not contingent on kin terms being compositional and can arise of its own accord. We also find that extremely high edit distance values can correspond with extremely high mutual information val-

ues. This suggests not only that predictive structure and compositionality need not co-occur, but also that they may interact to impose multiple kinds of structure on a kinship system simultaneously: predictive structure allows us to predict who is the descendant of whom, while compositionality encodes information about the shared features of individuals.

4. Study 3: does predictive structure emerge during generalisation?

What mechanisms give rise to the emergence of predictive structure in kinship terminology? Typological universals are evidence that language reaches "stable engineering solutions" that satisfy the demands of the processes of language learning and use by which languages are transmitted (Evans & Levinson, 2009). Indeed, typologically common structural properties of language such as noun-dependency ordering, (Culbertson et al., 2012; Culbertson, Schouwstra, & Kirby, 2020), harmonic ordering of phrasal heads (Culbertson, Franck, et al., 2020), and conditioned variation more generally (Hudson Kam & Newport, 2009; Smith & Wonnacott, 2010) have been shown to emerge by the imposition of cognitive biases associated with language learning. The typological prevalence of predictive structure in kinship terminology suggests a pressure for this structure that we argue is also imposed during learning.

Where might this pressure come from? We draw a parallel with the predictive structure of morphological paradigms (e.g. the organisation of tense or case inflections within a language), which vary their complexity along two dimensions: the number of different surface forms in the paradigm, and the predictive structure of the paradigm – how reliably you can predict a form from other cells in the paradigm (Ackerman & Malouf, 2013; Johnson et al., 2021). Acker-

man and Malouf (2013) note that while variability in surface forms seems to be quite unconstrained cross-linguistically, predictive structure seems to be much more constrained — even morphological paradigms with a great deal of superficial complexity are structured predictively, just as we find with paradigms of kinship terminology. Experimental work has shown that predictive structure licenses reliable generalisations about novel forms (Johnson, 2021); people are more likely to correctly infer the form for an unknown cell if it can be predicted using knowledge from another part of the paradigm.

Predictive structure in kinship terminology may be a consequence of this same generalisation process. If learners use their existing knowledge of kin terms to generalise to the terms for other kin, those forms are likely to be predictive of one another. For instance, when the *funcle* merger occurs (Figure 1), the *brousin* merger follows it because we take our existing information about funcles and generalise that category structure to their children, inferring that they too should be categorised together. The second merger takes hold because, when faced with incomplete input, learners will extend the terms they have learned on the basis of their existing knowledge - i.e. they will assume that the term *brousin* applies to all children of *funcles*. In short, kin terms evolve to suit the kinds of generalisations we are likely to make during the process of kin term acquisition.

To test whether predictive structure emerges during generalisation, we ran an artificial language experiment where participants generalised known kin terms to novel referents, taking inspiration from similar experiments where participants learn to generalise novel words in a hierarchical meaning space (e.g. F. Xu & Tenenbaum, 2007). Participants were shown family trees with some members labelled with a kin term and some left unlabelled. Their task was to assign a kin

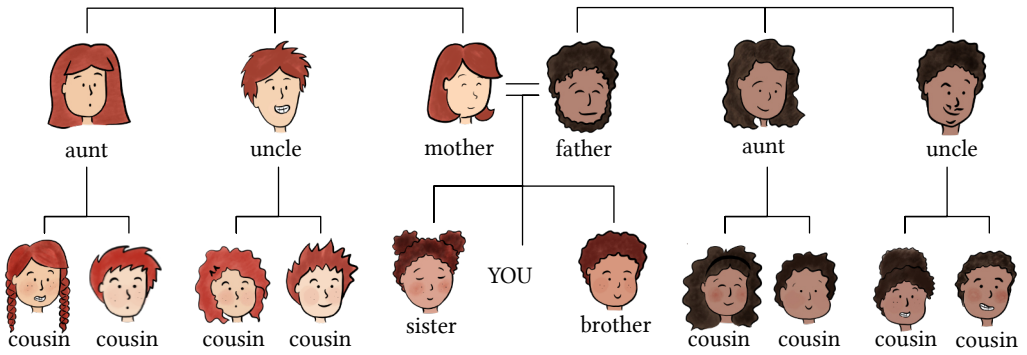


Figure 9: Example family tree diagram from our experiment, labelled here with English kin terms for clarity. Each tree comprised two generations of kin, including a mother, father, maternal/paternal aunt, maternal/paternal uncle in G+1 and a male sibling, female sibling, and a male and female child for each nuncle in G0. Participants saw a new tree with new characters in each of their eight trials.

term to each of the unlabelled referents. This paradigm let us test whether participants have a preference for predictive structure when generalising kin terms as we expected, or whether they generalise according to some other information, or whether they assign new terms haphazardly.

4.1. Methods

4.1.1. Participants

We recruited 40 participants from Prolific. Participants were self-reported native speakers of English resident either in the United Kingdom or the United States who had an approval rating of over 98% over all completed studies. We excluded three participants who failed at least two of eight attention checks, leaving 37 participants. The median completion time for the experiment was approximately 16 minutes, and participants were paid £2.86 for their participation.

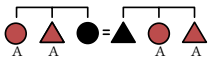
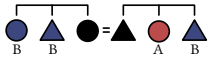
System	Description	Distinction	Tree structure
Type I	All parents' siblings share a term	None	
Type II_{side}	Parents' siblings are categorised into two groups of two	Side of the family	
Type II_{gen}		Gender	
Type II_{opp}		Opposite gender to parent	
Type III_{fb}	Parents' siblings are categorised into a group of three and a group of one	Father's brother distinct	
Type III_{fz}		Father's sister distinct	
Type III_{mb}		Mother's brother distinct	
Type III_{mz}		Mother's sister distinct	

Table 1: In each trial, participants saw one of eight different kinship systems. Kinship systems varied by which distinctions were present among G+1 kin. Type I systems have all parents' siblings categorised together with a single label. Type II systems have them categorised into two groups of two on the basis of gender (brothers vs sisters), side of the family (mother's siblings vs father's siblings), or relative to the connecting parent's gender (mother's brother and father's sister vs mother's sister and father's brother). Type III systems have them categorised into a group of three and a group of one (all combinations).

4.1.2. Design

We provided participants with a number of kin terms in artificial languages. Their task was to generalise from these known kin terms to new referents, using the distinctions made amongst parents and nuncle terms to infer distinctions amongst siblings and cousins.

In each trial, we presented participants with family tree diagrams comprising two generations of kin (see Figure 9). The family trees were populated with illus-

trated characters designed to share visual similarities; closely related kin shared features such as hair colour, hair style, and skin tone. All family trees were oriented with the parents and parents' siblings at the top with their children positioned beneath them. Participants were encouraged to imagine themselves at the centre of the tree; the word 'YOU' was written beneath the parents and between the siblings to indicate how they should imagine themselves to be related to everyone else in the tree. To avoid effects associated with the position of different relatives on screen, we varied the left-to-right order of men and women in the tree between participants, as well as whether the mother and father appeared on the right- or the left-hand side.

'Known' kin terms were given beneath the corresponding characters in the tree (see Figure 10). In each trial, participants were given all labels for G+1 kin, and also shown at least two labels in G0 so they could generalise these labels to the remaining unlabelled kin. The given G0 labels were randomly assigned in a way that was consistent with a high degree of predictive structure, but still left the remaining distinctions open to interpretation. We gave participants four labels to choose from for the remaining kin. Two of these label choices matched the provided G0 labels, and two were novel.[1] Participants could use as many or as few of the four labels as they liked.

Participants completed eight trials, each with a different underlying kinship

[1]We gave participants a choice of four labels after finding in a pilot version of the study that participants provided with only two labels commonly used those labels to distinguish gender only. With four labels, people did not have to choose between competing preferences: it was possible to distinguish gender as well as maximising the predictive structure of the system if they chose to.

system that differently distinguished parents' siblings.[2] These can be divided into three types on the basis of distinctions made among parents' siblings, given in Table 1. Varying the distinctions in the input ensured that our findings about predictive generalisation would be robust across different patterns of surface variation in kin term categorisation.

We designed eight sets of artificial kinship terms, so that participants would see different kin terms associated with each of the eight system types (see Table 2). These were generated by randomly concatenating syllables from a randomly selected inventory for each of the eight systems. Diacritics were added to each language to further distinguish each set of kin terms. Each set of kin terms was randomly paired with one of the eight system types at the start of the experiment, and the terms were randomly assigned to each category of relative; no two participants saw the same terms in the same orientation.

4.1.3. Procedure

The experiment was implemented with jsPsych (de Leeuw et al., 2023), and ran online in participants' web browsers.

In the task instructions, participants were told they were helping with a language documentation task, where intuitions about kinship terminology were needed to fill gaps in the vocabulary records of newly documented languages. They were told they would have to use the known terms in each language to in-

[2]The labels assigned to parents' siblings varied for each system, but parent terms always uniquely distinguished 'mother' and 'father'. This was done partially because it is cross-linguistically common for parental terms to be distinct, and primarily to avoid confusion for our English speaking participants who would not be used to parent terms being polysemous.

Language name	Labels
Zolawengï	*kïmy, qïru, bokokohï, zuvewu, mewaqïfe, morïky, ryvïri, lobowaby, bybï, hïlaki*
Pukepapu	*kawuka, goviku, lawolu, popingi, wupu, ngapapu, powisunge, lepego, luge, pygele*
Øwar	*aqinudød, øqoç, aweh, orinoçim, iciqusem, iqur, ømenad, adeçim, emøyeç, uquçun*
Aluwut	*azuf, egezalat, ozatof, otev, ulywaf, idanuluf, agivud, ilug, etupol, awupav*
Khêvkkêf	*kvurddåh, rhymsvåd, rhåv, ktot, tsåvvses, svik, ktåvrhus, smêvskêv, vrêd, sfêvdok*
Vpižodvipe	*vbāžo, rbāranžā, nžodo, pžicā, žbodu,vropcāre, vruži, ngcāžo, byure, cžunge*
Tuuquhaïbö	*šïïnö, cöahido, šaaqaude, nuubö, töeqo, tuumïhu, yaöpe, beiguumu, canšuehu, möïtu*
ăonuu	*taobyo, nyumuo, goozăi, qăo, niuhiê, baumiă, gêă, huanăa, nyumuo, hyonăo*

Table 2: Artificial label sets generated for this experiment. Participants saw different labels in each of their eight trials. In-keeping with the task framing that they were helping with language documentation in eight under-research languages, they were given the name of each artificial language. Labels for known kin were randomly chosen from the pool for each language, as were the labels participants could select from.

fer what the missing terms were likely to be, and that they would have a choice of four terms for each gap.

At the start of the experiment, all participants watched a short tutorial of how to interpret a family tree. Participants were asked to imagine themselves at the centre of the diagram, and characters appeared in a tree structure alongside a prompt describing their relationship in English; e.g. “This is your mother”. After completing the tutorial, participants moved on to the eight critical trials of the experiment, one for each system type. Each critical trial had an observation phase, where participants were exposed to the known kin terms for each language, and a selection phase, where participants had to label novel referents in the family tree (Figure 10).

In the observation phase, we introduced participants to the known kin terms one at a time: we showed them all the terms for G+1 relatives, and two labels

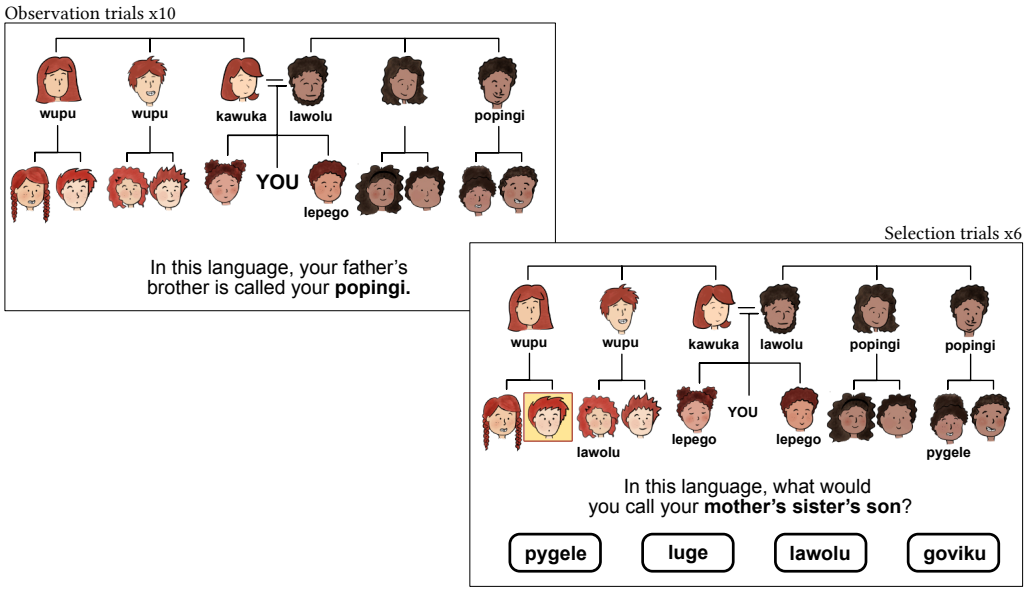


Figure 10: Left: Example observation trial for 'father's brother'. Participants saw each of the 10 known kin terms appear in the tree with a prompt telling them the term for that relative in an artificial language. Right: Example selection trial for 'mother's sister's son'. Kin who were labelled in the observation trial remained labelled. One by one, the six remaining kin were highlighted and participants were prompted to choose the label that best fit that referent. Once all kin were labelled, participants could choose to move on to the next trial.

for G0 relatives. The term would appear underneath a character with a prompt, e.g. "Your mother's brother is called your **popingi**". We chose to use English kin terms in the prompt to ensure participants understood that the label was a kin term and not a name.[3] Each term remained on screen beneath its referent after

[3]We do not expect English kin terms in the prompts to have primed any English-like behaviour. The nature of the task means that using an English-like categorisation is not actually possible: where sibling terms are given, they are not gender-distinct. And due to the given G0 labels, cousins must be categorised into at least two groups.

it was introduced, and participants could progress to the next term by clicking 'Next'.

Once all the known terms had been presented, the selection phase began. One by one, the unlabelled kin in the tree were highlighted and a prompt would appear, e.g. "In this language, what would you call your mother's sister's son?" Participants could then choose from four labels what they thought the best fitting kin term was for the referent. After their selection, that label would appear beneath the referent, and a new referent would be highlighted. They continued in this way until all kin were labelled, at which point they were asked to confirm their choices to move on to the next trial. Kin were highlighted for selection in a random order; this reduced the risk of any linear ordering effects and encouraged participants to pay attention to which person was being highlighted and how they were related to other individuals in the tree. Participants could undo their responses at any time, allowing them to backtrack on earlier labelling decisions, but they could not change the labels that were given to them originally. Once per trial, an already labelled individual would be highlighted and the participant was asked to correctly select their label as an attention check.

4.2. Results

Our hypothesis was that participants' generalisations would result in kinship systems with a high degree of predictive structure. We analysed the results from this study in two ways. First, we looked at participants' generalisation choices relative to the expectations under our hypothesis. Second, we measured the mutual information of participants' output languages and compared this to a simulated baseline distribution, analogous to the analysis of the typological data in Section 2. We also conducted some exploratory analysis to investigate which

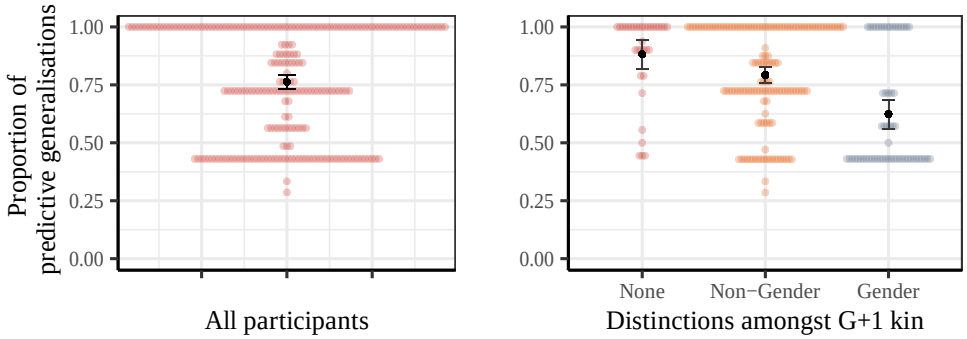


Figure 11: Proportion of predictive generalisations made by participants across all eight trials (left) and grouped by distinctions in the input (right). Black dots indicate group means, error bars indicate 95% confidence intervals. Generalising predictively is extremely common. However, 'clusters' of responses at 0.4 and 0.7 indicate that there may be other factors influencing participants' generalisations. Participants make fewer predictive generalisations when there is a gender distinction in their input compared to when there are no distinctions, or distinctions made on some other basis (e.g. side of the family).

kin are most frequently categorised together, and to test whether participants' responses were influenced by the distinctions made in their input.

4.2.1. Generalisation gives rise to predictive structure

For each label a participant assigned to a referent, we coded whether or not that generalisation was predictive or not — 1 if it increased the predictive structure of the system overall, 0 if it did not. This allowed us to measure how many generalisations from known kin terms to a new referent reflected a preferences for predictive structure. Figure 11 suggests that across the board, participants were very likely to generalise a known term in a way that increased predictive

structure. We performed an intercept-only mixed effects logistic regression[4], predicting predictive generalisations with an intercept and by-participant and by-system random intercepts, revealing that participants generalised predictively significantly more often than we would expect if they did not have a preference for maximising predictive structure ($b = 1.27$, $SE = 0.24$, $p < 0.001$).

But what were participants doing when they were not generalising predictively? Participants were highly systematic: in Figure 11 we see distinct 'bands' of responses, which indicate that many participants generalised in the same way. This tendency is particularly strong in systems that distinguish parents' siblings by gender (Type II_{gen} and Type II_{opp}), where only 23% and 29% of participants produce significantly predictive systems, respectively. Indeed, Figure 11 (right) suggests that participants are more likely to generalise predictively when there are no distinctions or non-gender based distinctions amongst G+1 kin, but have a much lower proportion of predictive structure where there are gender distinctions.

Do distinctions in the input affect generalisation behaviour? We fit a second mixed effects logistic regression, predicting whether or not a generalisation was predictive based on a fixed effect of distinction type (No Distinctions, Gender Distinctions, Non-Gender Distinctions) and by-participant random intercepts and random slopes for distinction type. Pairwise comparison between levels of the model revealed that participants were significantly less likely to generalise predictively when they saw Gender Distinctions compared to No Distinctions ($b =$ -1.676, $SE = 0.208$, $p < 0.001$) or Non-Gender Distinctions ($b =$ -0.929, $SE =$

[4]All models reported in this paper were run in R (R Core Team, 2024) using the packages `lme4` (Bates et al., 2015) and `lmerTest` (Kuznetsova et al., 2017).

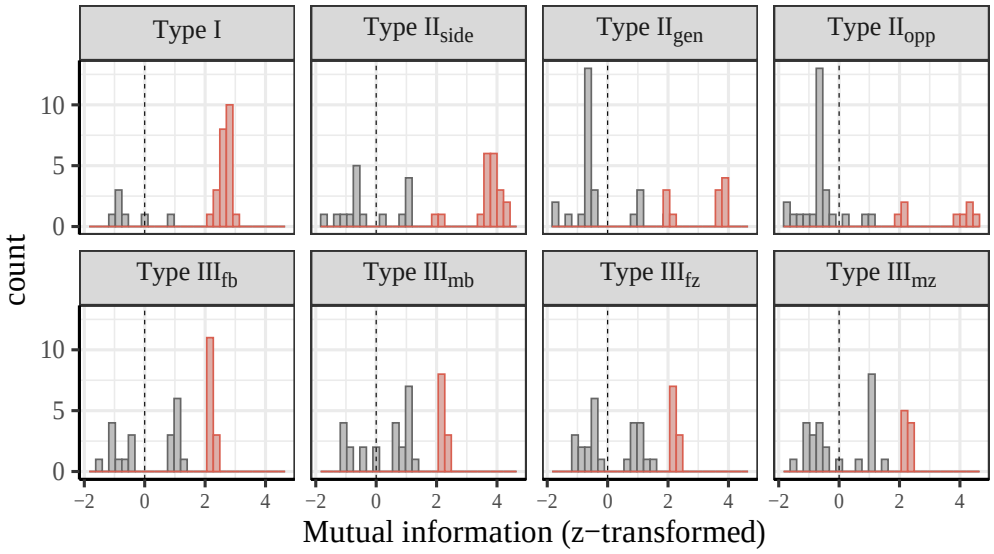


Figure 12: Distribution of mutual information z-scores for each system type (see Table 4.1.2). Red bars indicate participants who beat a significance threshold of $p < 0.05$. Dashed line shows $z = 0$. Participants produced kinship systems with an extremely high degree of predictive structure for all systems; note that 100% of participants who produced significant mutual information values produced the maximum possible mutual information for that system type. Performance is better for some systems (e.g. Type I) than others (e.g. Type II$_{gen}$ or Type II$_{opp}$), but participants never produce systems with significantly lower mutual information than the baseline.

0.134, $p < 0.001$). However, participants were also more likely to generalise predictively when they saw No Distinctions compared to Non-Gender Distinctions, albeit with a smaller effect ($b = 0.746$, $SE = 0.194$, $p < 0.001$). .

By analogy with the typological results described in Section 2.3, we also analysed the mutual information of the kinship systems produced by our participants. We expected that participants would produce kinship systems with higher mutual information than chance, indicating a preference for kinship terms with greater predictive structure. We compared each participants' responses to a null

baseline generated by the same method described in Section 2.2 for the typological data: each participant's responses were permuted 1000 times, randomising which kin terms were associated with which referents. Figure 12 shows each participants' z-transformed mutual information with respect to their bespoke baseline. Across all eight system types, 38% of participant-generated kinship systems have more extreme mutual information than expected under permutation (i.e. $z > 1.96$).[5] In addition, all 38% of the significant systems achieved the maximum possible mutual information for their respective system type, indicating that participants had a preference to structure kinship systems predictively.

The large number of participant-generated kinship systems that have extremely high mutual information supports our hypothesis that generalisation facilitates predictive structure in kinship terminology systems. But in the next section, we explore further how participants' approach to categorising kin varied depending on the distinctions made in their input.

4.2.2. Patterns of categorisation

Participants' responses were highly systematic, but they did not always generate kinship systems with a high degree of predictive structure. So, what kinds of distinctions were participants making when they were not generalising predictively?

Figure 13 represents the frequency with which referents were categorised

[5]This result may look less extreme than the typological data in Section 2, but it was harder to beat the baseline in this context: the kinship systems in this experiment had a maximum of 10 terms covering 16 possible meanings, and participants' choices could only affect the distribution of terms in G0. As a result, there were far fewer unique permutations for a given system than for the typological data.

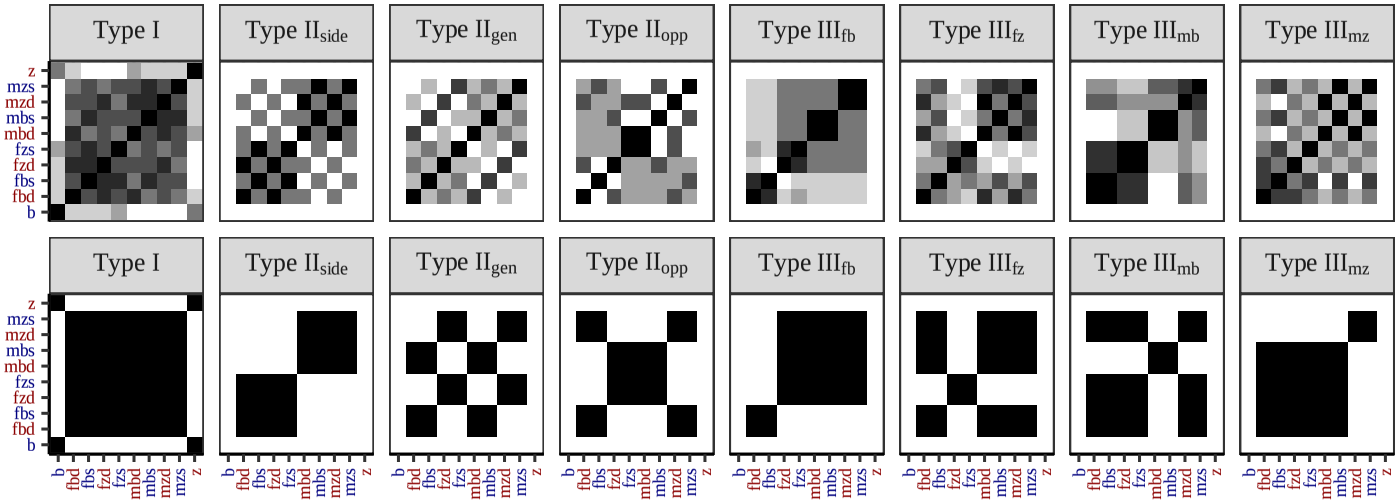


Figure 13: Frequency with which each pair of referents were categorised together for each system type. Top row: Participants' responses; Bottom row: ideal categorisation given maximally predictive structure. Each cell represents two referents, indicated on the axes - e.g. MZS = mother's sister's son, FBD = father's brother's daughter. Red axis text indicates female referents, blue axis text indicates male referents. Darker cells indicate that a greater proportion of participants gave two referents the same label. Diagonals are black, as each individual is always categorised with themselves. Similarities between the participants' responses (top row) and the ideal categorisations under our hypothesis (bottom row) indicate closer adherence to predictive structure, while deviations indicate that participants often generalised in a different way. These figures show that participants have a strong tendency to encode predictive structure, but the overlayed checkerboard pattern suggests that gender distinctions are also a strong driver of participant behaviour.

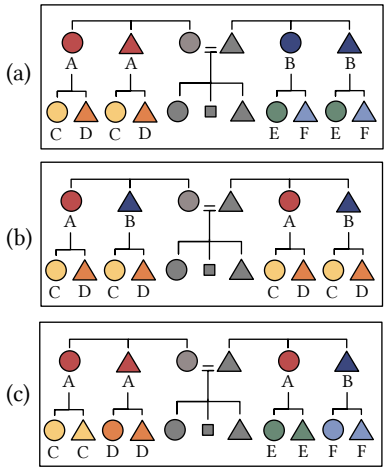


Figure 14: Schematics of participant responses across three system types. Tree conventions as in Figure 2. (a) A maximally predictive Type II_{side} system that also encodes gender among G0 kin. (b) A non- predictive Type II_{gen} system that only encodes gender among G0 kin. (c) A non-predictive Type III_{fb} system that treats each branch of cousins as a discrete unit.

together by our participants for each system, compared with the expected frequency assuming a preference for predictive structure. The greater the resemblance between each pair of plots, the more participants are producing predictive kinship systems. Participants produce predictive structure more consistently for some system types (e.g. Type I) than others (e.g. Type II_{opp}), corroborating our earlier results.

Across all systems, we see a strong preference to distinguish referents by gender. Referents are ordered male-female on the axes in Figure 13, so the 'checkerboard' appearance of the participants' responses indicates a preference to give separate terms to male and female referents. It was possible for participants to introduce a gender distinction *and* maximise predictive structure. We can see this behaviour in the plots for Type I and Type II_{side} where the participants' re-

sponses map closely to our expectations with the addition of the checkerboard pattern. Figure 14a gives an example of a participant response that maximises predictive structure and encodes gender.

However, the preference for gender distinctions overwhelms the preference for predictive structure in Type II_{gen} and Type II_{opp} systems. The ideal pattern has not emerged in the participants' responses, but the checkboard pattern is clear. Why would this occur? These systems encode gender distinctions among G+1 kin. The gender distinctions made in the input for these systems may have encouraged some participants to introduce gender distinctions in G0 at the expense of predictive structure. Figure 14b gives an example of a participant response that encodes gender, but does not have predictive structure.

The input data does not always condition participant response in a straightforward way. For Type III_{fb} systems, father's brother was distinguished from all other nuncles. While the participant plot for Type III_{fb} looks similar to the ideal plot, we see a pattern of darker cells on the diagonal. This suggests that participants prefer to treat each branch of the tree as discrete - each pair of cousins has a unique term (see Figure 14c) - rather than using the predictive solution of distinguishing father's brother's children from the other cousins.

The categorisation frequencies presented in Figure 13 suggest that the distinctions made in the input affect the distinctions participants will make. Predictive structure is one possible outcome of generalisation, but feature-based solutions like gender or lineage are also available.

4.2.3. Influence from gender distinctions in the given G0 labels

Our participants show a strong tendency to encode gender distinctions whether they are present in the G+1 terms or not. What else could be encouraging par-

ticipants to introduce gender distinctions?

Other than system type, the only part of the input that varied were the given G0 terms. These were randomly assigned per trial, so the gender of their referents varied between trials. With five possible female referents and five male, it was equally likely a participant saw two labelled G0 individuals of the same gender or of different genders in any given trial. When they saw two labelled G0 individuals of opposite genders, they may have interpreted this as evidence that the kinship system distinguishes individuals by gender, encouraging them to introduce gender distinctions.

We fit a mixed effects regression with fixed effects for whether or not participants generalised predictively and whether or not participants saw gender-distinct labelled G0 individuals. This analysis indicated a marginal tendency for participants to make fewer predictive generalisations when they saw gender distinctions in G0, but the effect was not significant (b = -0.2348, SE = 0.12, p = 0.062) indicating that gender distinction between the G0 individuals does not sway participants towards gender-based generalisation, unlike gender distinctions between the G+1 individuals. In the general discussion we consider extra-linguistic reasons why participants might prefer to encode gender.

4.3. Discussion

We found that participants are sensitive to predictive structure when generalising kinship terms to novel referents, and often produce kinship systems with a significant degree of predictive structure. These results are consistent with the typological prevalence of predictive structure in kinship systems that we reported in Section 2, suggesting that this universal emerged due to a preference for predictive structure imposed during the generalisation of known kinship

terms to unknown referents. Predictive structure is more prevalent in the world than in our experiment, but the miniature kinship systems used in our experiment have considerably fewer degrees of freedom than natural kinship systems. The fact that participants frequently generate kinship systems with the maximum possible mutual information is therefore extremely unlikely if there were no preference for predictive structure.

However, participants did not always maximise predictive structure during this task; they also have a preference to distinguish kin by means of gender or lineage. In our exploratory analysis we found that participants' responses could vary depending on the distinctions made in their input: gender distinctions amongst G+1 kin inspired generalisation of terms to same-gender individuals, while gender distinctions amongst G0 kin did not have this effect. While it is possible to have both gender distinctions and predictive structure, participants' behaviour suggests that the bias for maximising predictive structure can be outweighed in the presence of evidence for a different kind of categorisation.

5. General discussion

In this paper we set out to identify structural constraints on kin term diversity and the mechanisms that give rise to those constraints, in line with claims that kinship terminology systems are optimised for efficient communication (Kemp & Regier, 2012). Previous work has addressed either kin term diversity or mechanisms, but not both — by suggesting constraints that cannot explain the entirety of the cross-linguistic variation we can observe and why it emerged (Fox, 1979; Greenberg, 1990; Jones, 2010), or by offering mechanistic explanations based on experimental data obtained using artificial kinship systems that are quite re-

moved from what variation in natural language kinship systems actually looks like (Smith et al., 2020). Here, we have married these two approaches. We conducted the largest-scale typological study of kinship terminology to date and have shown that systems of kinship terminology near-universally exhibit predictive structure. Following this, we have shown that participants in the lab have a preference for predictive structure when generalising from known kin terms to new referents. We suggest that the prevalence of predictive structure cross-linguistically can be attributed to this preference.

Efficiency and kinship. Our results support the claim that systems of kinship terminology are optimised for efficient communication (Kemp & Regier, 2012). The constraint for predictive structure permits a lot of surface variation in the number of terms and what distinctions they make, while constraining the space of possible systems to those which are predictive. In Kemp and Regier's (2012) model, simplicity is partly measured by the compressibility of kin term definitions. We argue that predictive structure reflects a pressure for kinship systems to be simple, as kin terms are more compressible under predictive structure: to go back to the example in Figure 1, we can recursively use our definition for *funcle* 'father and his male sibling' to generate our definition for *brousin* 'the child of a *funcle*'. In other words, predictive structure facilitates simplicity and is orthogonal to expressivity - kin term meanings are simpler for being predictive, and so the cost of having more expressive, granular terms is reduced.

Simpler, more compressible linguistic structures are associated with ease of learning, while more expressive structures are associated with pressure to encode distinctions (Kirby et al., 2015). We propose that predictive structure emerges during generalisation: part of the language learning process where some knowl-

edge of the system is available (and learned), but some innovation is still required (Holtz, 2024). When the learner is faced with an inductive problem, generalising from known terms to new referents who have not yet been assigned a kin term in the mental lexicon, they do not innovate at random but use their existing knowledge of the system to label the new referent in a systematic way. The internal co-selection process of kin term evolution (Passmore et al., 2021) is facilitated by this process of generalisation: kinship systems evolve to be more generalisable.

Emergence of feature distinctions. In our experiment, generalisation did not always yield predictive structure. Our participants often generalised other properties of the system, such as whether it encoded distinctions by gender or by lineage. Our exploratory analysis suggests that in some cases, gender distinctions in the input kin terms increase the chances of participants encoding gender distinctions between the novel referents. Elsewhere, feature distinctions emerge without a clear priming effect from the input. This suggests there may be several biases influencing participants' behaviour at this task: one that encourages producing kinship systems with predictive structure and one that encourages the repetition of feature distinctions. These are not by definition at odds with one another: we see many participants produce kinship systems with both predictive structure *and* gender distinctions. Future work could explore how feature distinctions and predictive structure interact in the cross-linguistic data.

Innovating distinctions by gender was also common when there were no gender distinctions elsewhere in the system (other than between mother and father). What could encourage this behaviour? Our participants are native English speakers, which could explain their willingness to encode gender distinctions due to the *brother/sister* and *aunt/uncle* distinction in English. However, English

does not have a gender distinction for cousins, and yet many of our participants distinguished parents' siblings children by gender.

There are a number of extralinguistic reasons why our participants may have introduced additional distinctions not present in their input. Kin term distinctions in the native language are deeply ingrained to the extent that they modulate performance at cognitive tasks and affect what features of individuals are salient (Anggoro & Gentner, 2003). The characters in our experiment are varied in appearance and look most similar to their immediate relatives, but gender may be the most salient visual characteristic by which to distinguish individuals (at least for our participants). Additionally, models of social discrimination suggest that social categories like gender (and ultimately, gender bias) emerge robustly with minimal pressure to do so (O'Connor, 2019), suggesting that distinguishing individuals by gender may be a readily available hypothesis when faced with this task. Some participants also introduced a branch distinction, grouping each pair of cousins together as a discrete unit, particularly for Type III systems. These are indicated by visual characteristics like hair or skin colour, which may also be salient visual cues for our participants. Additionally, native English speakers are used to bilateral systems of descent, where you perceive yourself to be related equally to your mother and father's sides of the family. They are more familiar with symmetry rather than lineage as a structuring element of kinship terms: in English, distinctions are symmetrical on each side of the family tree. This may have made our participants reluctant to introduce asymmetrical distinctions (where e.g. mother's siblings and their children are distinguished at a finer grain than father's siblings and their children), even where this would be the most predictive solution for Type III systems. It is possible that respondents

from unilateral societies (e.g., Hindi speakers, who have additional distinctions between father's siblings compared to mother's siblings) might respond differently.

However, culturally-motivated biases and visual cues do not form the entire basis of participants' behaviour, as predictive structure is not culturally motivated nor visually salient, yet it frequently emerges.

Kinship terminology and social behaviour. In this paper we have treated kinship terminology primarily as a linguistic phenomenon, and predictive structure as a universal shaped entirely by biases associated with language transmission. We have shown that structural properties of kinship are associated with efficient communication, but the social dimension of kinship is also relevant to how we communicate about kin. Therefore, it is important to point out that measures of communicative efficiency for kinship do take into account that the need to refer to particular kin varies cross-culturally (Kemp & Regier, 2012).

Within anthropology, it has long been accepted that patterns of kinship terminology reflect patterns of social structure, and that this is a driving force in kin term variation (Murdock, 1949). For instance, tolerance to cross-cousin marriage may be associated with terminology that distinguishes marriageable kin from non-marriageable kin (Viveiros de Castro, 1998). Recently, with the availability of large cross-linguistic datasets, we see that correlations between kin term organisation and social practice may be confined to particular language families or regions, and may not be as robust as previously thought (Passmore & Jordan, 2020; Passmore et al., 2023). Nonetheless, even if the link between terminology and behaviour is not straightforward, future work should consider the interplay between efficient communication and social practice, by taking advantage of the

large linguistic and cultural datasets now available to inform cognitive scientific models. Accounts of kin term organisation should be aware of how kin distinctions are shaped by the repeated imposition of culturally-motivated communicative need, determining how often we communicate about particular individuals, who we perceive to be similar, and what distinctions we need to make.

That said, our experimental design offers a new paradigm for robustly testing claims about kin term diversity that are *removed* from participants' own cultural frames of reference. Artificial language experiments in general allow us to test linguistic claims in a language- and culture-neutral way. In introducing this task framing, we hope to provide a new avenue for testing claims about kinship and cognition, particularly in the light of new contributions to cognitive science that use kinship terminology as case study to develop models of lexical acquisition and semantic refinement (e.g. Mollica & Piantadosi, 2022; Markham et al., 2024).

6. Conclusion

In this paper, we presented a large-scale typological analysis of kinship terminology, showing that the world's languages exhibit an extreme degree of predictive structure between generations of kin terms. We also conducted an artificial language experiment, finding that participants often generalise known kin terms to new referents in a way that maximises predictive structure. We suggest that the cross-linguistic tendency for kinship systems to have predictive structure emerges due to pressures imposed during cultural transmission: when individuals are faced with incomplete input when learning kin terms, they will predict the remaining kin terms based on their existing knowledge, resulting in highly structured systems.

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Appendix A. Data quality affects mutual information score

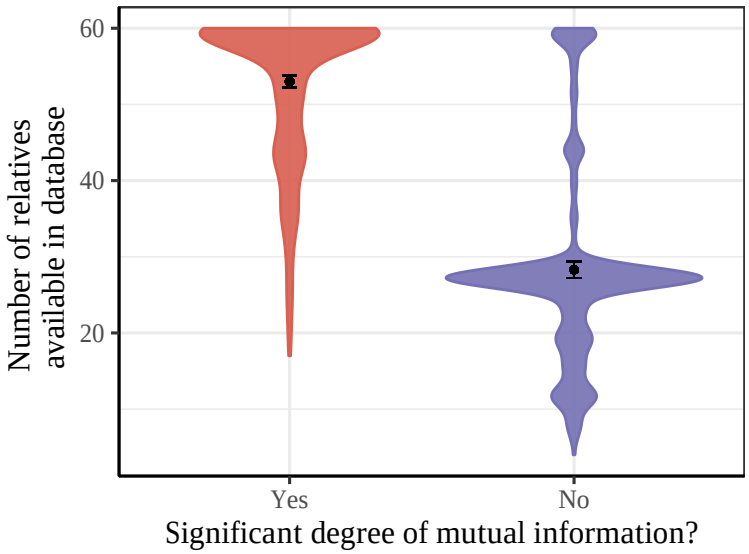


Figure A.15: Amount of data available in Kinbank for languages which had extreme values of mutual information when compared to their baseline (significance threshold $p < 0.05$). The maximum number of relatives who could be listed in the database was 60 (across G0 and G+1). On the whole, languages with insignificant values had less data available, suggesting that this may have been in part an artefact of data quality.

Some languages in our sample had non-significant degrees of mutual information than we would expect if there were no preference for predictive structure. Here, we show that this is largely a problem with data quality. Not all languages in Kinbank have complete data for the G+1 and G0 relatives we used for this analysis. If a language is missing data, then not only is its veridical mutual information value limited (as more data results in a larger value), but also its baseline will be composed of fewer permutations, making it harder for the veridical value to surpass the baseline. Figure A.15 shows that languages whose mutual infor-

mation score was *not* significant (i.e. a z-score less than 1.96 and greater than -1.96) tended to have fewer relatives in their dataset. The maximum number of relatives was 60.